

Syllabus for

Academic year 2021/2022 >

FFM234 - Vector fields and classical physics

Vektorfält och klassisk fysik

Syllabus adopted 2021-02-08 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

4,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57124

Open for exchange students: No

Maximum participants: 130

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0117 Examination 3,0 c Grading: TH	3,0 c						Sat 23/10-2021 am J, Mon 03/01-2022 am J, Mon 15/08-2022 pm J
0217 Project 1,5 c Grading: UG	1,5 c						

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Tünde-Maria Fülöp](#)



[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

MVE035 Calculus of Several Variables.

TMA671 Linear Algebra and Numerical Analysis

Aim

The aim of the course is to provide the student with a basic knowledge of mathematical methods used in the description and the analysis of physical phenomena in three-dimensional space, in addition to deepen her or his understanding of fundamental classical physics.

Learning outcomes (after completion of the course the student should be able to)

After successfully completing the course the student will be able to analyze problems in physics and engineering science that involve spatial variations of physical observables. Specifically, the student will have acquired basic mathematical tools for differentiation and integration of scalar and vector fields analysis and synthesis of fields, including singularities.

The student will be able to give an account of the field concept in classical physics, and apply it to elementary problems in the theories of classical electrodynamics (Maxwell's equations) and thermodynamics (diffusion equation and the equation for heat transport).

After having completed the course the student may continue with more advanced studies in classical physics.

Content

The field concept in classical physics; scalar and vector fields. Differentiation and integration of fields. Sources, sinks, and vortices. Superposition, analysis and synthesis of fields. Scalar and vector potentials. Solutions of field equations; boundary value problems. Elementary applications in the theories of classical electrodynamics and thermodynamics.

Organisation

The course is based on a series of lectures and exercise classes. The course includes a compulsory project on computer-assisted methods.

Literature

- "En första kurs i matematisk fysik", Martin Cederwall (available for download)
- "Mathematical methods for physicists" (7th ed), by Arfken, Weber and Harris, 2013 [In particular, parts from chapters 1-10]
- Additional material will be announced on the course web page.

Examination including compulsory elements

- Compulsory project on computer-assisted methods.
- Written exam at the end of the course.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.